

ReactJS HOL- 9

Objective:

Develop a React application named cricketapp demonstrating ES6 features including map(), arrow functions, destructuring, spread operator and conditional rendering.

Theory

Features of ES6

ES6 introduced let, const, arrow functions, classes, template literals, destructuring, modules, promises, map, set and the spread operator.

JavaScript let

let declares block-scoped variables that can be reassigned.

Difference between var and let

var is function scoped and can be redeclared; let is block scoped and cannot be redeclared in the same scope.

JavaScript const

const declares block-scoped constants that cannot be reassigned.

ES6 Class Fundamentals

Classes provide constructors, methods and object-oriented programming support.

ES6 Class Inheritance

Inheritance uses the extends keyword to reuse properties and methods.

Arrow Functions

Arrow functions provide concise syntax and preserve lexical this.

Set and Map

Set stores unique values. Map stores key-value pairs preserving insertion order.

Implementation Steps

1. Create React project: `npx create-react-app cricketapp`
2. Create `ListOfPlayers` and `IndianPlayers` components.
3. Use `map()` to display player names and scores.
4. Use arrow functions to filter scores below 70.
5. Use destructuring for odd and even players.
6. Merge T20 and Ranji Trophy arrays using the spread operator.
7. Render components conditionally using a flag.
8. Run using `npm start`.

ListOfPlayers.js

```
const
players=[{name:'Jack',score:50},{name:'Michael',score:70},{name:'
John',score:40},{name:'Am',score:61},{name:'Sachin',score:95},{na
me:'Dhoni',score:100}];
export default function ListOfPlayers(){
  const below70=players.filter(p=>p.score<70);
  return(<div><h1>List of Players</h1>{players.map((p,i)=><li
key={i}>>{p.name} {p.score}</li>)}<h2>Scores Less than
70</h2>{below70.map((p,i)=><li key={i}>>{p.name}
{p.score}</li>)}</div>);
}
```

IndianPlayers.js

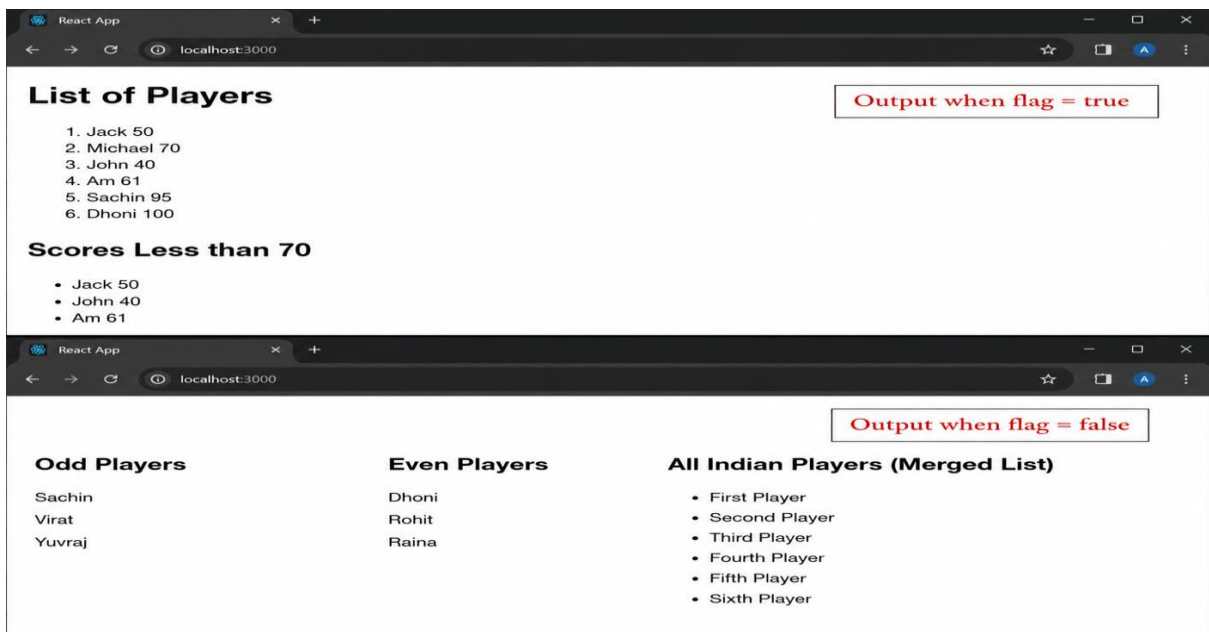
```
const team=['Sachin','Dhoni','Virat','Rohit','Yuvraj','Raina'];
export const
OddPlayers=()=>{const[first,,third,,fifth]=team;return<div><p>{fi
rst}</p><p>{third}</p><p>{fifth}</p></div>}
export const
EvenPlayers=()=>{const[,second,,fourth,,sixth]=team;return<div><p>
```

```
>{second}</p><p>{fourth}</p><p>{sixth}</p></div>
const T20=['First Player','Second Player','Third Player'];
const Ranji=['Fourth Player','Fifth Player','Sixth Player'];
export const IndianPlayers=[...T20,...Ranji];
```

App.js

```
import ListOfPlayers from './ListOfPlayers';
import {OddPlayers,EvenPlayers,IndianPlayers} from
'./IndianPlayers';
const flag=true;
function App(){
  return
  flag?<ListOfPlayers/>:<div><OddPlayers/><EvenPlayers/>{IndianPlay
ers.map((p,i)=><li key={i}>{p}</li>)}</div>;
}
export default App;
```

Output Screenshot:



Learning Outcome

Implemented ES6 features including map(), filter(), arrow functions, destructuring, spread operator and conditional rendering in React.

